

Robert Brownstein

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Experience

Gnarled Helix LLC | Founder

(2022 - present)

- Building a 2D platformer where the player learns JavaScript to solve puzzles and defeat enemies.
- Custom engine includes TypeScript, React, Three.js rendering, Vite, Rapier2D physics, RTK state management, and JS-Interpreter based sandboxed javascript runtime.
- Bootstrapped team of 10, launching on Steam with Electron packaging in Q3 2026.

Boston Dynamics | Staff Software Engineer

(2022 - 2026)

- Built major features for Orbit, the mission scheduling, teleoperation, and data review solution for Spot, the quadrupedal industrial inspection robot. Notable feature work included real-time robot monitoring, map editing, mission editing, inspection editing, and data visualization improvements. TypeScript frontend (React, Three.js), TypeScript and C++ on the backend. Protobuf data handling, Bazel build system.
- Built robot-side features relating to thermal panoramic image capture, networking, API connectivity, and maintenance in TypeScript / Python / C++.

Better | Senior Software Engineer

(2019 - 2022)

- Improved core workflow engine and a variety of API integrations with our many external partners.
- Introduced GraphQL into our core product, and built automatic type converters to automatically generate and sync types from TypeScript.
- Various UI work with React, Webpack, Rollup, Ember.

Social Tables (now Cvent) | Engineer -> Engineering Manager -> Lead Engineer

(2016 - 2019)

- Lead development of layout automation API with novel algorithm, authentication and permissions infrastructure, external developer portal, 3D scan to floorplan converter.
- Managed a team of 4 engineers that built and released multiple products, including a data-intensive user dashboard and an internal floorplan authoring tool (partially released open-source as [OpenFPC](#)).
- Resumed work as an IC in order to work on challenging front-end engineering problems in Diagram, the company's flagship in-browser CAD product. Built advanced template management, section management, 3D preview capabilities and optimized core 2D rendering engine.

Hinge | Backend Engineer

(2015)

- Maintained and improved a Python based service layer powering a mobile app with 0.5M DAU.
- Introduced CI into development workflow, and tackled scaling challenges while modernizing a legacy codebase and DB schema with over a 1B records.

Education

BS in Computer Science - University of Maryland, College Park

(2014)

Skills

- **Major languages:** JavaScript / TypeScript, Python, C++.
- **Favorite Libraries:** React, Redux, Three.js, Express, Protobuf.
- **Build and Deployment Systems:** Bazel, Webpack, Rollup, Github Actions.
- **Other Tooling:** Docker, AWS, Postgres, RethinkDB, ElasticSearch.